

אוניברסיטת תל אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר לתארים מתקדמים ע"ש זנדמן-סליינר

חימום מיקרוגל ממוקד של בזלת –

התכה ופליטה של לבה ושל פלזמה מאובקת

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בהנדסת חשמל ואלקטרוניקה על ידי

יואב שושני

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Melting, lava-like eruption, and dusty-plasma ejection**

A thesis submitted toward the degree of
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This research was carried out in The School of Electrical
Engineering- Department of Physical Electronics,

under the supervision of

Prof. Eli Jerby

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תקציר

חימום מיקרוגל ממוקד (Localized microwave-heating LMH) והיווצרות של נקודה חמה בתהליך של בריחה תרמית עלולים לגרום להיתוך מקומי, ואף לפליטה של פלזמה על ידי חומרים שונים. תופעה זו, הנוצרת על ידי אינטראקציה מקומית של קרינת מיקרוגל עם חומר, מאפשרת חימום מהיר בתוך אזור מוגבל (נקודה חמה, Hotspot) שממדיו קטנים בהרבה מאורך הגל. תהליך חימום מקומי זה מתפתח עד להופעת מעבר פאזה של החומר המוצק, לנוזל, גז או פלזמה. התנאים הדרושים להיווצרות נקודה חמה תלויים בהספק מקור המיקרוגל, וכן בגיאומטריה של המוליך ובמאפייני החומר המוקרנים.

עבודה זו מציגה מחקר ניסיוני ותיאורטי של אינטרקציה החימום הממוקד בין בזלת למיקרוגל. בעבודה זו אנו חוקרים לראשונה תופעה של פליטת פלזמה מאובקת מאבן בזלת מותכת, וכן היווצרות עמודות אש ופלסמוידים (plasmoids) דמויי כדור אש. בפרט, אנו מציינים את הדמיון בין השפעות LMH שונות באבני הבזלת (שנצפו בקנה מידה מעבדתי) לבין תופעות וולקניות ענקיות בטבע, כגון התפרצות וזרימה של לבה, פליטה של ברק-כדורי (Ball lightning) ויצירת אבק וולקני.

מערך הניסוי שלנו כולל מהוד מיקרוגל בנפח של $\sim 10^3 \text{ cm}^3$ המעורר בהספק של 1kW. בתוך המהוד מונחת אבן בזלת בנפח של $\sim 30 \text{ cm}^3$. בתהליך LMH, הליבה הפנימית של אבן הבזלת מתחממת ומותכת תחילה. הקרנה נוספת מובילה להתפרצות בזלת מותכת מליבת האבן אל פני השטח (בדומה להתפרצות הר געש ופליטה של לבה מתוכו). הבזלת המותכת מתקררת ומתמצקת ליזוכית וולקנית שחורה (Obsidian). במחקר זה מצאנו כי הקרנה נוספת גורמת לבזלת לפלוט סילון של פלזמה מאובקת, המתפתחת לעמוד פלסמה, ולאחר מכן לפלסמויד הדומה לברק-כדורי כפי שנצפה לעתים רחוקות בטבע.

עמודי-אש אלו והפלסמוידים בצורת כדורי-אש מאופיינים בניסויים באמצעים שונים, תוך כדי התהליך ולאחר סיומו. טמפרטורות העירור והסיבוב של הפלסמה בהם (T_{exc} , T_{rot}) מוערכות בטווח של 0.3-0.6 eV, עם צפיפות אלקטרונים ואבק של $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$ ו- $n_d \sim 10^{16} \text{ m}^{-3}$, בהתאמה. תכולת היסודות המרכיבים את הפלזמה המאובקת, המתגלים על ידי פליטת הספקטרום האטומי שלה, נמצאה דומה לתכולתה של אבן הבזלת ממנה נפלטת הפלזמה, וכן דומה לתצפיות ספקטרליות של ברק-כדור טבעי. שכבת האבק שנצברה על גבי המשטח המאסף (בקצב צבירה של $\sim 0.2 \text{ mm}^3/\text{s}$) מכילה מיקרו-ננו-חלקיקים במבנים שונים. אבקה זו, שהופקה על ידי הפלזמה המאובקת, מאופיינת בהרכב יסודות הדומה לאבן הבזלת הפולטת, וכן לאפר וולקני.

תוצאות הניסויים נידונות בהיבטים מדעיים ומעשיים שונים, כגון הרלוונטיות שלהן לתופעת הברק הכדורי ותופעות וולקניות נוספות וכן ביחס ליישומים שונים, כגון ייצור אבקה, ספקטרוסקופיה באמצעות מיקרוגל (Microwave-induced breakdown spectroscopy MIBS), ומיצוי של מינרלים מאבני בזלת.

Abstract

Localized microwave-heating (LMH) and the consequent formation of hotspots by thermal-runaway instability may cause local melting, and even plasma ejection by various materials. This phenomenon, generated by the microwave-material interaction, enables a rapid heating within a localized zone (a hotspot, much smaller than the microwave's wavelength). This local heating process evolves up to the occurrence of a phase transition of the solid material into liquid, gas or plasma. The necessary conditions for the hotspot formation depend on the microwave power source, the geometry of the applicator, and the irradiated material properties. This work presents experimental and numerical investigations of the LMH interaction of basalts and microwaves.

In this work we first study a phenomenon of dusty-plasma ejection from molten basalt, forming fire-column and fireball-like plasmoids. We also indicate the resemblance of various LMH effects in basalts (observed in laboratory scales) to the gigantic volcanic phenomena in nature, such as lava eruption and flow, ball-lightning emission, and creation of volcanic dust.

Our experimental setup employs a microwave cavity of a $\sim 10^3\text{-cm}^3$ volume, powered by a 1-kW magnetron. A basalt stone of a $\sim 30\text{-cm}^3$ volume is situated in the cavity. By applying LMH, the inner core of the basalt is first melted. Further irradiation leads to the eruption of the molten basalt, from the core to the surface (similarly to lava eruption and flow in volcanoes). The molten basalt cools down and solidifies to a black glass (obsidian). Here we show that further irradiation causes the basalt to emit a plume of dusty plasma, which evolves to a confined plasma column, and further to a plasmoid similar to a natural ball-lightning. These fire-column and fireball-like plasmoids are characterized by various *in-* and *ex-situ* means. Their excitation and rotational temperatures are estimated in the range $\sim 0.3\text{-}0.6$ eV, with electron and dust densities of $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$ and $n_d \sim 10^{16} \text{ m}^{-3}$, respectively.

The elemental content of the dusty-plasma, revealed by its atomic spectral emission, is found similar to the content of the originating basalt, and to spectral observations of natural ball-lightning as well. The dust layer accumulated on the cavity's upper collector (in a deposition rate of $\sim 0.2 \text{ mm}^3/\text{s}$) contains micro- and nano-

particles in various structures. This powder, deposited by the dusty plasma generated in the cavity, possess an elemental composition similar to the emitting basalt stone, and to natural volcanic ash as well.

The experimental results are discussed in various scientific and practical aspects. These include their potential relevance to natural ball-lightning and volcanic effects, as well as their significance for several applications, such as powder generation, microwave-induced breakdown spectroscopy (MIBS), and mineral extraction.

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The measurements of material properties by means of SEM, EDS and XRD in Chapter 3 were initiated by Dr. Zahava Barkay and Dr. David Levi from Tel-Aviv University. This research is supported by The Israel Science Foundation (Grant No. 1896/16

Finally, I would like to thank my family for their love and support that made this work possible.

Dedication

I am dedicating this thesis to the memory of my grandmother, Rivka Segal R.I.P, who was always curious about science and research, and showed great interest in this work progress and publications.

This thesis is also dedicated to the memory of my grandfather, Ezra Shoshani R.I.P, who always showed great respect and admiration for academic achievements.

Both passed away during the last months.

Your memory will light my way and will ever be with me.

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1. Introduction

1.1. About basalts

Basalts, the most common volcanic rocks on Earth, appear in nature in a range of physical and chemical properties, and in a variety of outer shapes [1]. Basalt underlies more of Earth's surface than any other type of rock, and is the key component of oceanic crust. Above sea level, basalt is common in hotspot islands and around volcanic arcs, especially those on thin crust. The largest volumes of basalt on land correspond to continental flood basalt, which are known to exist in India, Canada, Brazil, Russia, South Africa, USA, etc. It is also common on the moon and on other rocky planets in the solar system. The dark areas visible on Earth's moon, the *Lunar Maria* are plains of flood basaltic lava flows. Basalt commonly erupts on Io (Jupiter's third largest moon), and has also formed on Mars, Venus and the asteroid Vesta [2].

Basalt is used for a wide variety of purposes. It is most commonly crushed for use as an aggregate in construction projects, such as road base, aggregates for concrete and for asphalt pavement, railroad ballast, filter stone in drain fields, and many other purposes. Basalt is also cut into dimension stone. Thin slabs are cut and sometimes polished for use as floor tiles, building veneer, monuments, and other stone objects. Additional use for basalt is fiber reinforcement, in which thousands of fibers are compressed, and wound spirally into a structural reinforcement rod [3]. The basalt rebar is corrosion and rust-resistant, it was originally viewed as an ideal material for use in marine construction and on overpasses and bridges which were often exposed to salt.

Basalts are common aphanitic igneous extrusive (volcanic) fine grained rocks. It is composed of minute grains of plagioclase feldspar (generally labradorite), pyroxene, olivine, biotite and hornblende [4]. Basalts are formed by the rapid cooling of basaltic lava from the interior of the Earth mantle and exposed at, or very close to, the surface of the Earth. Significant amounts of basalt are produced above oceanic hotspots. These are locations where a small plume of hot rock rises up through the mantle from a hotspot on Earth's core. The Hawaiian Islands are an example of where basaltic volcanoes have been built above an oceanic hotspot, Kilauea (had a series of eruptions this year, 2018) is the most active of these volcanoes. It has been in an almost continuous eruption since January 1983. Basalt flows from Kilauea have extruded over one cubic mile of lava,

which currently covers about 48 square miles of land. These flows have travelled over seven miles to reach the ocean, covering highways, homes and entire country side areas that were in their path.

More than 500 million people live within the potential exposure range of a volcano, according to Doocy *et al* [5], there were a total of 91,789 deaths, 14,068 injuries, and 4.72 million people affected by volcanic events between 1900 and 2008. Volcanic eruptions not only produce direct material losses; they can also affect global climate thus affecting society as a whole, even a small eruption, as occurred in Iceland of mount Eyjafjallajökull in 2010, caused major air traffic disruptions over northern Europe causing disastrous socio-economic impacts [6]. According to forecasts by scientists at the *Geological Society of London* [7], agriculture would be impossible in many volcanic-eruption affected regions. Furthermore, crops will even be destroyed by a layer of ash just 1-cm thick. Etna, the biggest volcano in Europe, caused enormous damage when it erupted in 2001 and again in 2002/03. In 2002, the ash rain alone caused economic losses of around 800 m€ [8].

In the past year (2018) we have experienced several hazardous volcanic eruptions, leading to massive number of casualties, evacuation of populations and damage to houses, property and infrastructure. The violent eruption of Guatemala's Fuego volcano [9] on June 3, 2018 [10,11], Indonesia's Mount Merapi volcano in a series of eruptions from mid-May for several weeks, Telica volcano in Nicaragua on June 18, 2018 [11], and Hawaii's Kilauea volcano eruption on May 4 [12], all accompanied by earthquakes, fissures, ash plumes, flying rocks and pyroclastic flows, fast-moving currents of extremely hot gas and volcanic matter that pour down volcanic slopes at incredibly high speeds [13]. As seen in the latest Kilauea eruption, pyroclastic flow is the most dangerous volcanic activity, ongoing for several weeks, and forcing thousands of people to evacuate their homes.

Volcanoes directly or indirectly produce, or host, deposits of aluminium, diamonds, gold, nickel, lead, zinc, and copper. Intensely altered and fractured basalt is a common host rock for gold. Diamonds are brought to the surface from the mantle in a rare type of magma called Kimberlite, and erupted at a rare type of volcanic vent called a diatreme or pipe. Some South African mines operate at 25 carats of diamond per 100 cubic meters of rock, or about 2 grams of diamonds per 100 tons of rock.

Basalt drilling is used for geo-thermal wells, tunnelling and mining as basalt being an abundant rock in the Earth crust. Mechanical drilling of basalt and other hard rock has been proven to be the most economical and hence most commonly used one. During basalt breakage in mining and civil applications, the performance of mechanical equipment such as tunnel boring machines (TBM's) is limited by high levels of bit wear and maintenance. Therefore, methods of preconditioning, to weaken natural rock prior to mechanical breakage (for example, flame torch-assisted TBM's [14]) has been researched in the last decades.

1.2. Microwave processing of basalts

The concept of using microwave energy to generate heat for rock fracture and drilling was briefly examined in the 1960's [15], but due to technical issues at the time, it was not deemed economical and was not further investigated, till the 2000's [16].

Heating, cracking, braking and melting of basalt rock by means of microwave irradiation have been studied in the last decades as microwave technology has advanced. Basalt microwave-heating experiments were presented in various aspects. Melting of basalts in a microwave cavity was introduced in 2004 [17]. More recently, microwave heating of basalt at lower temperatures was studied in various conditions [18]. The formation of cracks due to temperature gradients and thermal stresses after microwave heating in a cavity was analysed by a thermo-mechanical model [19], point load and strength measurements were taken [20]. Rock breakage and cracking by means of horn antenna microwave irradiation was conducted, testing the temperature and strength of the basalt rocks after different irradiation powers and distances from the antenna [21]. All studies showed melting of the basalt rock after a period of time depending on the irradiation power. Larger basalt stones, of $\sim 400 \text{ cm}^3$, were heated by microwaves to $\sim 500^\circ\text{C}$ in their core [22]. In 2013, Jerby *et al* showed that basalt melting by localized microwave heating (LMH) involves a thermal-runaway instability leading to a hotspot formation in the core of the basalt stone causing a volcano like effect of core melting, cracks and lava flow from the core towards the surface [23].

1.3. Microwave heating

Microwave heating of materials is a unique technology for industrial processing, widely studied and implemented [24-26]. The absorption of microwave energy allows volumetric heating of materials thus it leads to time saving and reduces energy consumption comparing to conventional heating. The microwave heating may comprise of several distinctive mechanisms such as ohmic, dielectric and magnetic losses. In conventional heating, the temperature gradient transfers heat from the surface towards the bulk via heat conduction. In contrast, in microwave volumetric heating the temperature gradient plays an opposite role, it directs heat outward from the bulk thus tends to cool it down through heat convection from the surface. Since heat transfer is a diffusive process by nature, the typical heating rate of a thick sample by means of conventional heating might be slower than by microwave volumetric heating.

Microwave heating of materials is often preformed volumetrically in a resonant cavity with transverse dimensions comparable to the microwave wavelength λ_0 employed. The electromagnetic (EM) energy is then dissipated into heat within the matter [27].

Microwave volumetric heating is limited to non-metallic materials and to metal powders. In these cases, the more lossy the material is (in the microwave range), the more drastically the energy decays within it, hence the less it penetrates into the bulk's core.

Microwave power applications are nowadays wide-spread throughout industry, science, and domestic households. The best known is thermal processing of foods, but one should also note other industrial, scientific, and medical (ISM) applications, such as drying of textiles, fibers, paper and board, material processing and sintering, and pre-heating and curing of plastics and rubber.

1.4. Localized microwave heating (LMH)

Microwave-heating techniques used in industrial and domestic applications have a unique feature of inner-body heating [28]. The hotspot effect (undesired in most microwave applications) may evolve due to thermal-runaway instability [29]. This tends to occur in materials characterized by properties with certain temperature dependencies (e.g. if the dielectric losses increase and/or the thermal conductivity decreases as the

temperature rises). Then, the hotter part of the material (in the core) tends to absorb relatively more microwave energy than the cooler part (on the surface). This positive-feedback effect induces localized microwave-heating (LMH) and thermal-energy concentration, which creates a molten hotspot inside the material core [30].

While most of the known microwave-processing techniques are volumetric in nature, the microwave-drill is an example for an intentionally localized applicator [16, 31]. The underlying physical effect that enables its sub-wavelength focusing of the microwave energy is the induced thermal-runaway mechanism [32]. This effect enables for instance the focusing of 2.45-GHz microwaves (12-cm wavelength) within the material into a $\sim 1\text{-mm}^{\varnothing}$ spot ($\sim 10^{-2}$ smaller than the wavelength). This unique capability of LMH enables for instance melting, drilling and other localized material-processing applications usually performed by lasers. On the other hand, unlike the remote laser operation, the localized microwave concentration is a near-field effect which requires a closed proximity (or even a physical contact) of the microwave applicator and the processed object. LMH effects were demonstrated by microwave-drill type applicators in various materials, including concrete, glass, ceramics, basalts, and bones [17,33-35].

1.5. Microwave-induced plasma ejection from solid materials

In the 1950's, Kapitza [36, 37] produced fireballs using high-power radio waves at atmospheric pressures. He suggested accordingly an external electromagnetic-energy mechanism for the production of ball-lightning in nature. The excitations of laboratory ball-lightning-like effects have also been demonstrated experimentally using high-voltage discharges on silicon [38-40] and in water [41]. Though various characteristics of ball lightning have been demonstrated experimentally, also by microwave-generated fireballs, a definitive experimental demonstration of a long lived, floating ball-lightning-like object has yet to be achieved.

Plasma-columns, plasmoids and fireballs, directly excited by localized microwaves from solid substrates to air atmosphere, were first presented in 2006 by Dikhtyar and Jerby [42] and have been studied since in various experiments [43-48]. A molten hotspot is created on the substrate's surface in a thermal-runaway instability caused by the localized microwave heating (LMH). This leads to a plasma-column ejection from the hotspot, and to oxidation or combustion processes within the plasma.

The plasmoid may be further evolved to form a self-sustained fireball while its feeding plasma-column is detached from the molten hotspot. Measurements of the optical emission spectra of such plasmoids show the dominance of elements imported from their originating solid substrates (either dielectric or metallic). These microwave-generated fireballs float in the air, hence demonstrating to some extent also the buoyancy feature of ball lightning.

Using synchrotron radiation, the small-angle X-ray scattering (SAXS) method was applied as well to study fireballs of this type, and to find their particle size distribution, density and decay rate. SAXS results of fireballs ejected from borosilicate glass [43] show that these fireballs contain particles with diameters of ~ 50 nm and average number densities of $\sim 5 \times 10^9 \text{ cm}^{-3}$. Hence, these fireballs are considered a form of dusty plasma [49-51] consisting of ensembles of charged nanoparticles in the plasma volume.

The creation of artificial and natural fireballs was studied also by Bychkov [52] as originating from a reservoir made up of an oxide cover filled with metal powder. As in natural ball lightning, it is electrically charged due to the transfer of charges from the linear lightning. A plasmon resonance theory [53] proposes that an object irradiated by EM waves may visually manifest itself as ball lightning. The plasmon resonance effect may cause also a strong concentration of EM-energy near particles, as observed experimentally in optical wavelengths.

A dusty (or complex) plasma is a normal electron-ion plasma with an additional charged component of micron-sized particulates. Complex (dusty) plasmas occur for instance in candle flame, the zodiacal light, in lightning in thunderstorms containing smoke-contaminated air, in volcanic eruptions, and in ball lightning. The dust particles in the plasma are charged by thermionic emission from the heated dust grains.

Thermionic emission involving plasma plumes are seen in the process of laser ablation when laser pulses are emitted on a metal surface [54], the ablation process is dominated by heat conduction, melting, evaporation and plasma formation [55]. The ablation of metals involves a complex sequence of different processes. One of the first is the thermionic emission process, followed by plasma plume ejection from the heated area. Surface electron emission due to thermionic and photoelectric effect contribute to plasma initiation [56].

1.6. The ball-lightning enigma

The ball-lightning enigma has attracted the interest of both the public and scientists [57-59]. It is yet a major motivation for studies of microwave-excited fireballs, which mimic ball lightning. This rare natural phenomenon has been accidentally observed by eyewitnesses, mostly during thunderstorms, earthquakes, volcanic activity, and tornadoes. These sightings describe ball lightning as glowing luminescent objects floating in the air and lasting for up to several seconds. Ball-lightning effects were observed in various shapes, bouncing on the ground, rotating as tornados, moving against winds, and even passing through glass windows.

A model for ball-lightning generation in nature, proposed by Abrahamson and Dinniss [60,61] suggests that normal lightning striking the ground evaporates silica and carbon, and ejects a plume of nanoparticles into the atmosphere. The reduction of silicon dioxide by carbon to silicon nanoparticles, in reactions such as



and their oxidation again, generate the heat energy that sustains the reaction. The ball-lightning luminosity is maintained by a slow combustion process in which the silicon nanoparticles are gradually fused together to form larger silicon spheres with oxidized outer surfaces. The oxide layer slows the further oxidation of the interior due to the higher melting point of the oxide, hence delaying the process by reducing further air oxygen diffusion to the silicon cores of the particles. The ball lightning is then evolved as an elastic aerogel structure because of mutual electrostatic aggregation of bipolar charged nanoparticles. Therefore, according to this model, the ball lightning seems to be a hybrid of plasma and combustion effects.

Ball-lightning and fireballs effects have been long associated with volcanic eruptions and volcanic air pollution. A large number of fireball sightings has been recorded in the last centuries [62], many occurring near volcanos or at times of heavy air pollution caused by volcanic eruptions. Spectrograph measurements of natural ball-lightning reveals the presence of Si, Fe, Ca and Al [63], as in soil, in line with the theory suggesting that ball lightning is generated by lightning striking the soil [64].

1.7. Research objectives and methodology

In this thesis, we have investigated various LMH-basalt interactions using a commercial 1kW magnetron at 2.45 GHz. This work advances previous studies performed by our group, including thermal-runaway effects in basalt-microwave interactions (such as core melting, lava eruption and microwave-drilling of basalt) [17, 23]. This work is also related to other studies of LMH interactions with various materials and metals, such as silicon, titanium and copper [42-48]. These studies presented dusty-plasma ejections from a molten hotspot in the irradiated material in the form of fire-column and fire-ball.

The thesis objectives were also to validate the basalt-LMH theoretical model by experiments and numerical simulations. A hypothesis of plasma ejection from basalt hotspots was proposed, and means for hotspot formation and plasma ejection were studied. The integration of an electrode and later an array of electrodes in the experimental setup was conducted in order to examine ways to expedite and intensify the LMH process in basalts. The effect of the basalt rock shape on the melting process and plasma ejection was studied as well. Another objective was to characterize the dusty-plasma ejected from basalt.

The thesis consists of an experimental work and theoretical analyses of the different setups, including diagnostics of the heating and melting processes (and comparison to the theoretical model), and characterization of the dusty-plasma ejected from the basalt samples. The plasma temperature, the electromagnetic properties, and the elemental composition, were measured by several *in-* and *ex-situ* means. Also, characterization of the dust accumulated from the dusty-plasma is presented in this study.

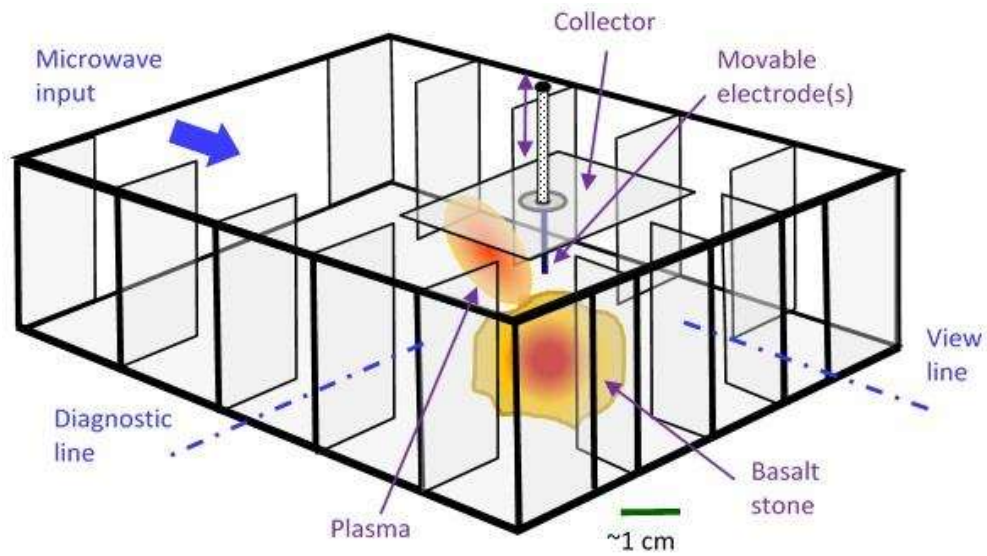
2. Experimental setup and procedures

2.1. The LMH furnace

The experimental setup shown in Fig. 2.1 consists of a microwave cavity fed by a 2.45-GHz, 1-kW magnetron. The cavity is made of a 10x5 cm² cross-section waveguide, shorted at the end by a mirror made of metallic vanes under cutoff. This arrangement enables a direct view into the waveguide, with no disturbance to the various diagnostic means (e.g. the thermal camera which requires a direct line of sight). A similar array of vanes under cutoff is also installed as the side wall of the cavity, enabling a broader view (from both perpendicular directions) and more in-situ diagnostic means simultaneously. As shown in the block diagram (Fig. 2.1b), the magnetron's switched-mode power supply (MagDrive-1000, Dipolar Ltd., Sweden) provides an adjustable input microwave power up to 1 kW. The microwave power is delivered to the cavity via an isolator and an impedance auto-tuner (Homer, S-Team Ltd., Bratislava, Slovak republic). The incident and reflected waves are recorded by the impedance-analyser mode of the auto-tuner. This enables adaptive impedance matching and optimal transmission of the microwave power into the cavity. The diagnostic means include video recording (up to 200 fps) and thermal imaging by FLIR-SC300 camera. The optical spectra emitted from the lava and plasma eruptions are captured by an optical spectrometer (Avaspec-3648) with 0.3-nm resolution in the wavelength range of 200-1000 nm.

A LabView script was written for each component of the setup to synchronize operation, acquire data and write time-stamps. The control screen offers: (a) Magnetron output power control via MagDrive programming, (b) HOMER definition control, measured and de-embedded reflection coefficient (magnitude and phase) data acquisition, (c) video-camera recordings control, and (d) voltage measurements operation and acquisition for the Langmuir probe measurements (Appendix A).

(a)



(b)

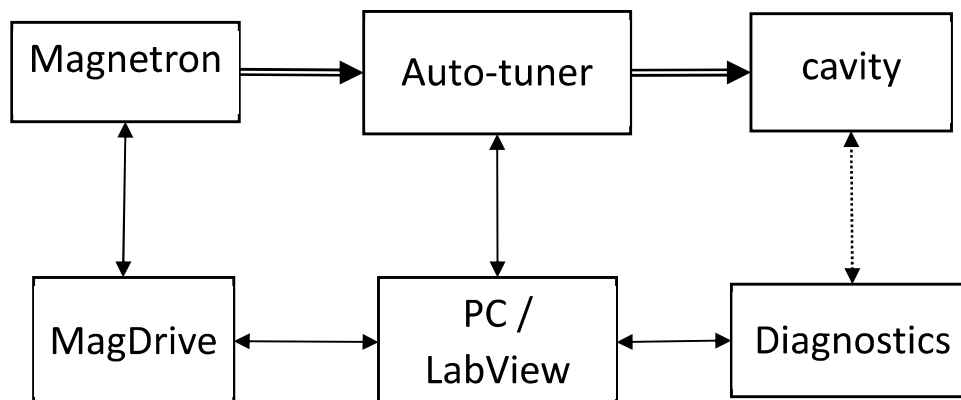


Figure 2.1: (a) A schematic of the experimental setup: chamber, movable electrode and basalt sample.(b) A block diagram of the experimental instrumentation, with the diagnostics including an optical spectrometer, thermal and video cameras, and an I - V probe.

2.2. Electrodes

In this experiment we use basalt stones (from the Golan heights) either in their original natural shapes or cut to (3-cm^3) cubic bricks. The basalt-rock-load is placed inside the cavity in an optimized position (discussed in Section 4.3) whereas the movable electrode directs the microwave energy locally into its substrate. Various levels of focusing the microwave energy are optionally available in this setup, either by moving a single electrode shown in Fig. 2.1a, or by manipulating an array of 9 or 18 electrodes (Fig. 2.2a and b respectively) installed in the cavity's ceiling via the collector, and in the cavity floor. Each electrode in this array expedites the intentional excitation of the hotspot in its vicinity, when brought into contact with the basalt-stone surface, and hence enables the stimulation of the fire-column (and consequently of a fireball) in a more controlled fashion.

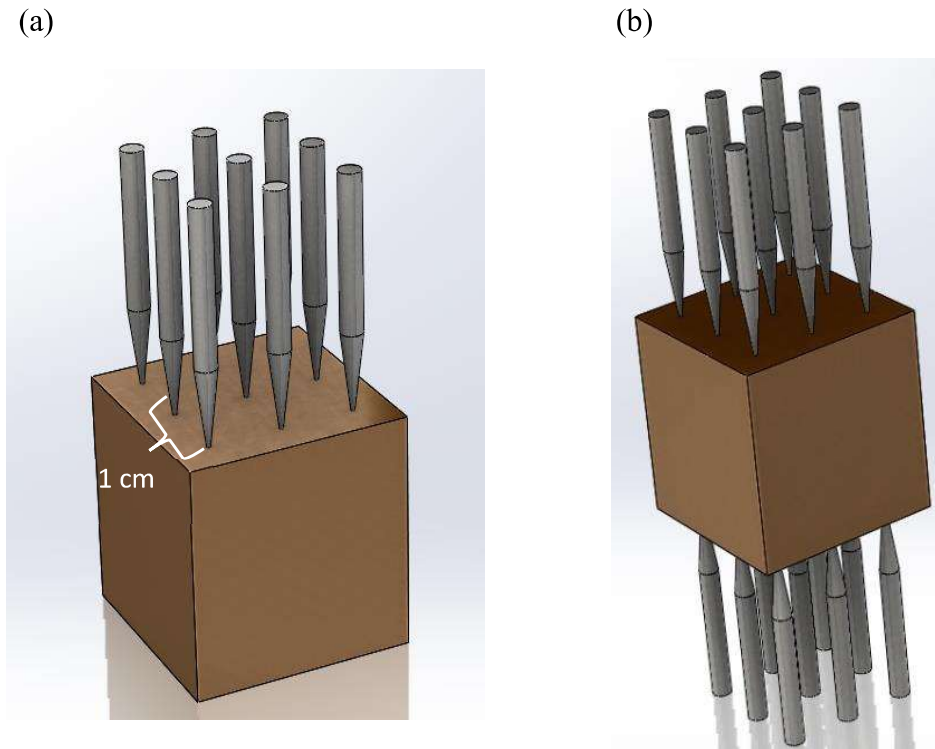


Figure 2.2: Electrode arrays. (a) An array of 9 electrodes pushed against the basalt surface.
(b) An array of 18 electrodes, with 9 electrodes on each face.

2.3. Diagnostic means and procedures

The video camera recording is used to monitor, measure times, and synchronize between the various effects observed (e.g. inner hotspot formation, cracks, breakage, lava eruption and plasma ejection) and the other diagnostic means used in the experiment. Once ejected, the plasma column size, shape and position can be found from the video recording. Also, the evolution of fireballs, as well as their shapes and movements, are monitored by the video camera.

The FLIR-SC300 thermal camera monitors the spatial temperature evolution of the sample, hotspot formation, temperature and size. Recordings and analyses are made by FLIR-Tools+TM software. Profile temperatures, temperature evolution in time, the hotspot size and temperature evolution, are all monitored and later analysed and synchronized with other experimental measurements. The thermal imaging data is later compared to simulation in order to validate the physical model and the simulation results.

2.3.1. Microwave scattering, reflection and transmission analyses

The HOMER impedance tuner (by S-Team Ltd.) is controlled as part of the LabViewTM system. Adaptive tuning is applied to intensify the hotspot heating in order to eject the plasma. The synchronized de-embedded reflection coefficient (S_{11}) measurements are presented and stored for offline analyses. In order to get more accurate measurements, once the plasma is ejected, one can cancel the adaptive tuning action in order to measure the actual reflection coefficient instead of the de-embedded one. This is made possible due to the self-tuning effect of the plasmoid (later discussed in Section 3.2.1), which provides an alternative to the adaptive impedance tuning by the impedance tuner.

The plasma transmission coefficient (S_{21}) measurements were carried out by a Tektronix-TTR500 series vector network analyser (VNA). The two-port system was constructed of two excitation ports with the plasma column in-between (Fig. 2.3a). For this purpose, two electrodes are placed alongside the plasma column and serve as the excitation ports. The plasma can be characterized by measuring changes in the coupling between the two electrodes using a VNA for the S -parameter measurements. At frequencies higher than 3 GHz, the S -parameter measurements are too noisy and thus it is difficult to notice changes in the coupling. A different approach for high frequency